Attachment 3
Wastewater Pretreatment Plant (WWPP) Chemical Treatment Process
Plating Wastewater Pretreatment Plant (PWPP)

PROCESS DESCRIPTION: In the Plating Wastewater Pretreatment Plant (PWPP) plating wastewater is treated by removing the heavy metals from the process wastewater. The wastewater is collected in holding tank T-26 (equalization tank in PWPP) from the two plating operations including T-4A (surge tank in Annex Basement RM 23-2-1A). (**NOTE: T-4A is the beginning of the PWPP**. All equipment from T-4A is covered under this contract, including all the piping system to T-26 from the Annex building to the main building.) The wastewater is then pumped to T-4 and then pumped to an oil/water separator. The wastewater then flows into treatment tank T-1 where sodium bisulfite is added to reduce hexavalent chromium to trivalent chromium. 93% sulfuric acid is also added to T-1 to maintain a pH less than 3.0 to reduce the required residence time in T-1. The wastewater then flows to tank T-2 where the pH is adjusted with 50% sodium hydroxide. The wastewater next flows through a mellifier (clarifier) for gravity separation of precipitated solids, and the supernatant wastewater flows to a surge tank T-5. Polymer from tank T-9 is also added to the upstream of the mellifier at an in-line mixer, as needed, to assist in the separation/precipitation process. From tank T-5, the water is pumped through a filter media system to tank T-6, and then to the sewer. Water from tank T-6 is also used to back wash the filter media. The dirty backwash water is pumped to T-4, and the water is then slowly returned to tank T-1 for treatment. The sludge is pumped from the mellifier to a sludge storage and dewatering tank T-10. The sludge in T-10 is pumped to a sludge filter press where the sludge is concentrated into sludge cakes for disposal. The water, which is discharged from the press, flows to the sump beneath the filter press T-25 and is pumped back to T-2 for reprocessing. The solids from the filter press are disposed of in accordance with EPA regulations.

PROCESS OPERATION: The Contractor shall operate the PWPP with prudent engineering and operating practices. Operating the PWPP includes, ***but is not limited to***, the following tasks:

(1) The Contractor shall take readings every two (2) hours of the pumps, flows, tank levels, data set points and operating conditions from the PLC. It should be noted that in the future this may be at least partially performed by SCADA software.

(2) The Contractor shall perform an inspection of the entire PWPP, every two (2) hours minimum during operation or more often as needed for proper operation.

(3) The Contractor shall take jar test samples of the wastewater to monitor for settling as necessary for the proper operation of the PWPP.

(4) The Contractor shall check the mellifier, and solids pumped to the filter press as needed.

(5) The Contractor shall start, stop, empty the filter press, prepare the drums, install the liners in the drums, package the sludge, fill each drum, secure

the lid, and properly label the drums. The drums for the sludge will be BEP provided and must be UN certified.

(6) The Contractor must obtain COR approval before sending any Hazardous Waste containers out of the Plant.

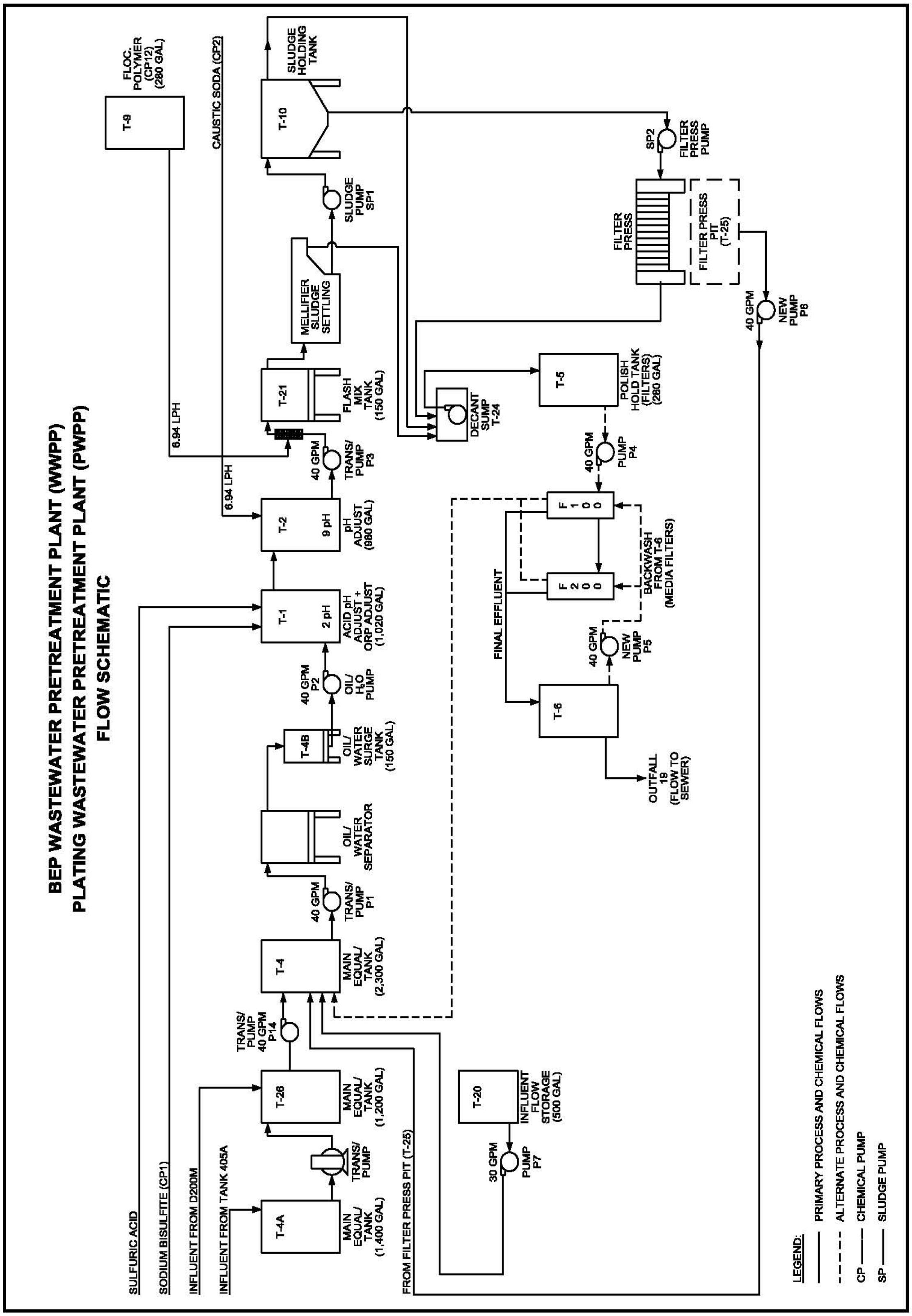
(7) The Contractor shall mix or replenish the treatment chemicals as needed.

(8) The Contractor shall check pH set points and local set points and calibrate pH and ORP meters weekly at a minimum or more often, as needed.

(9) The Contractor shall respond to all alarms, and correct the conditions to assure proper operation of PWPP is maintained.

(10) The Contractor shall inspect and clean the tanks at least bi-annually (twice per year) or more often as needed.

(11) The Contractor shall order the materials/supplies as needed to assure continuous operation of the PWPP.


BEP WASTEWATER PRETREATMENT PLANT (WWPP)
PLATING WASTEWATER PRETREATMENT PLANT (PWPP)
FLOW SCHEMATIC
SULFURIC ACID
SODIUM BISULFITE (CP1)
INFLUENT FROM D200M
INFLUENT FROM TANK 405A
FROM FILTER PRESS PIT (T-25)
T-4A
MAIN EQUAL/ TANK (1,400 GAL)
TRANS/ PUMP
T-26
MAIN EQUAL/ TANK (1,200 GAL)
TRANS/ PUMP 40 GPM P14
T-4
MAIN EQUAL/ TANK (2,300 GAL)
40 GPM
TRANS/ PUMP P1
OIL/ WATER SEPARATOR
T-4B
OIL/ WATER SURGE TANK (150 GAL)
40 GPM P2
OIL/ H2O PUMP
T-1
2 pH
ACID pH ADJUST + ORP ADJUST (1,020 GAL)
T-2
9 pH
pH ADJUST (980 GAL)
6.94 LPH
40 GPM
TRANS/ PUMP P3
T-21
FLASH MIX TANK (150 GAL)
MELLIFIER SLUDGE SETTLING
SLUDGE PUMP SP1
T-10
SLUDGE HOLDING TANK
T-9
FLOC. POLYMER (CP12) (280 GAL)
CAUSTIC SODA (CP2)
6.94 LPH
SP2
FILTER PRESS PUMP
FILTER PRESS
FILTER PRESS PIT (T-25)
40 GPM
NEW PUMP P8
DECANT SUMP T-24
T-5
POLISH HOLD TANK (FILTERS) (280 GAL)
40 GPM
PUMP P4
F 100
F 200
BACKWASH FROM T-6 (MEDIA FILTERS)
FINAL EFFLUENT
40 GPM
NEW PUMP P5
T-6
OUTFALL 19 (FLOW TO SEWER)
T-20
INFLUENT FLOW STORAGE (500 GAL)
30 GPM
PUMP P7
LEGEND:
PRIMARY PROCESS AND CHEMICAL FLOWS
ALTERNATE PROCESS AND CHEMICAL FLOWS
CP CHEMICAL PUMP
SP SLUDGE PUMP